Binary Heart Disease Prediction Using Machine Learning

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Dataset Overview and EDA

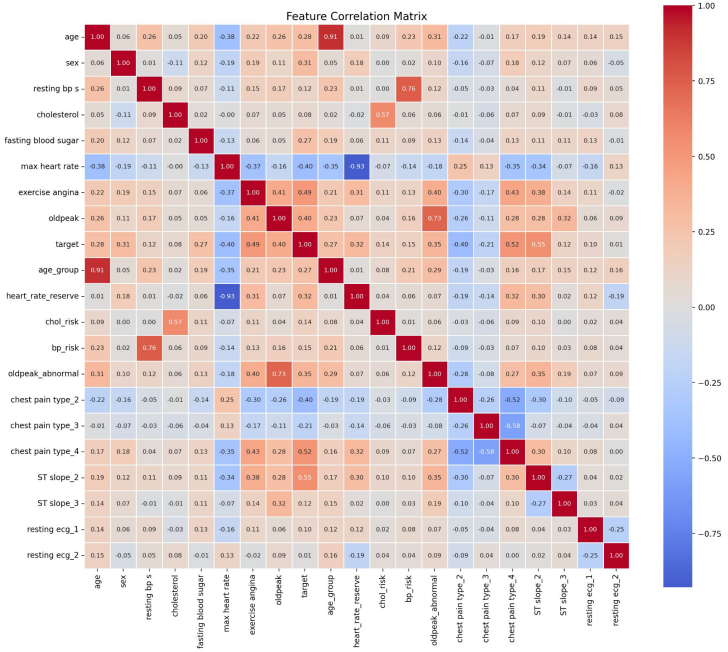
- Source: [Public Kaggle Heart Disease Dataset]
- Original size: 1190 samples

—> After cleaning: 918 samples —> 80/20 split

- Features: 11 clinical and physical input features
- Target variable: 1 = Heart Disease;

0 = No Disease

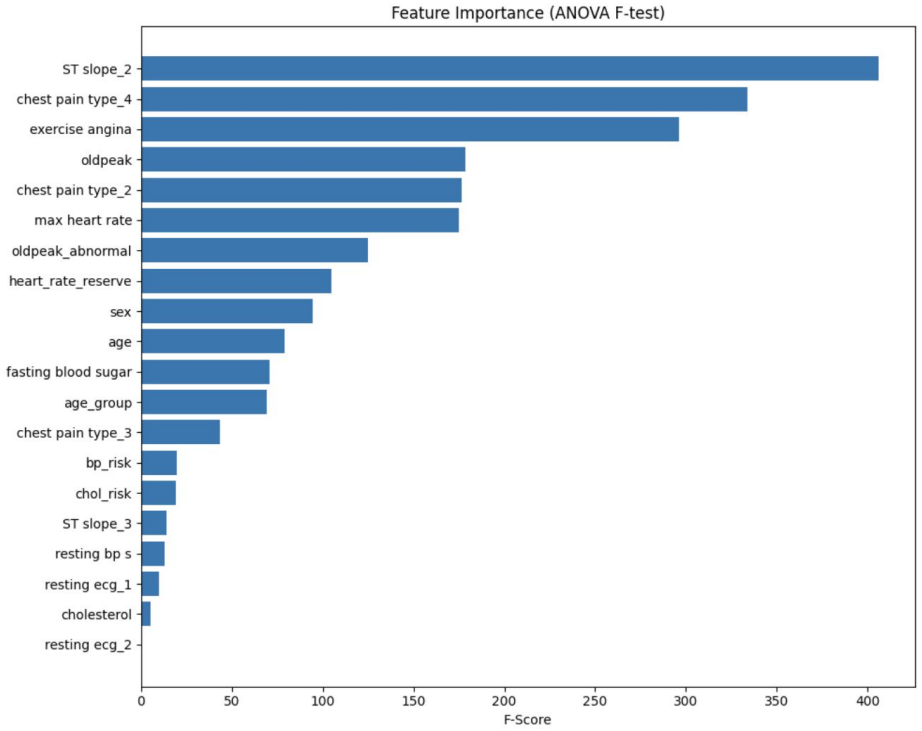
- Class distribution: 508 positive + 410 negative

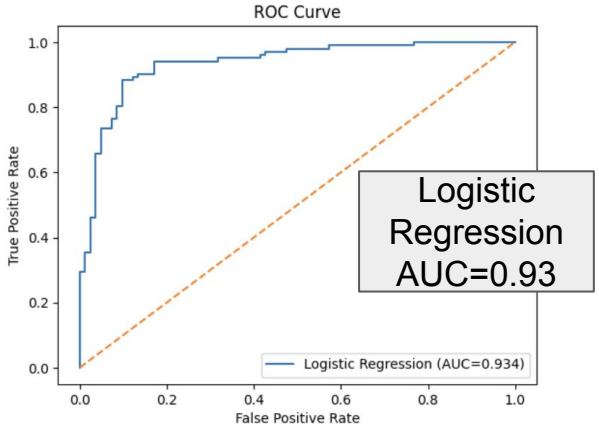
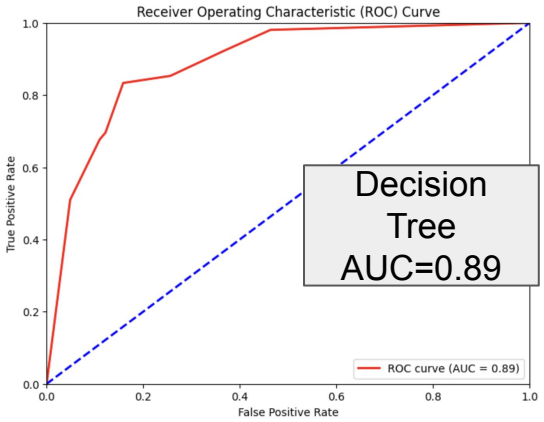
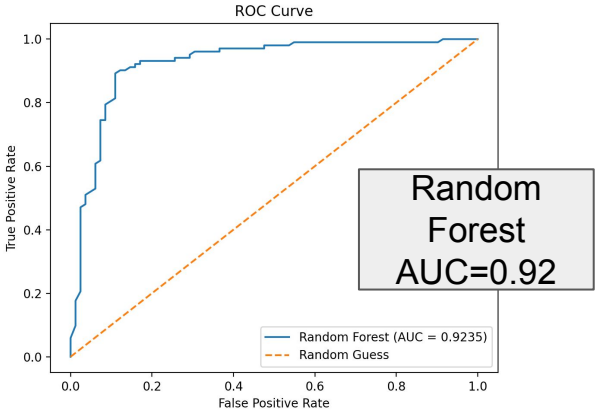


Feature Engineering

Key engineered feature:

- age_group
- heart_rate_reserve
- chol_risk
- bp_risk
- oldpeak_abnormal



Modeling and Evaluation Framework

	Model	Accuracy	Precision	Recall	f1-score
Baseline	Logistic Regression	0.875	0.91	0.86	0.88
Non-linear	Decision Tree	0.837	0.87	0.83	0.85
Advanced	Random Forest	0.8913	0.9020	0.9020	0.9020